Comprehensive Analysis and Dietary Strategies

1.INTRODUCTION:

This project analyzes the dietary habits, lifestyle, and food choices of college students using Tableau.It aims to identify patterns in nutrition, eating behavior, and health awareness through interactive dashboards.The analysis helps understand how students' food preferences influence their overall health and well-being.Data visualization enables quick and effective interpretation of dietary trends for informed decision-making.

Healthy eating habits play a vital role in maintaining physical health, mental well-being, and academic performance among college students. However, due to busy schedules, academic pressure, and lifestyle changes, many students adopt unhealthy dietary practices.This project focuses on analyzing the food choices and dietary habits of college students using Tableau, an interactive data visualization tool. The study examines factors such as meal frequency, dietary preferences, BMI, exercise habits, nutritional awareness, and food consumption patterns.

1.1 Project Overview:

Comprehensive Analysis and Dietary Strategies with Tableau – A College Food Choices Case Study is a data analytics project that examines the dietary habits and food choices of college students using interactive Tableau dashboards.
The project collects and analyzes data related to dietary preferences, meal frequency, Body Mass Index (BMI), exercise habits, nutritional awareness, food consumption patterns, and lifestyle behaviors to understand their impact on students' overall health.
Using Tableau, the collected data is transformed into meaningful visualizations such as bar charts, pie charts, line charts, heat maps, and interactive dashboards, making complex information easy to interpret.

The insights generated from this analysis support data-driven decision-making and help educational institutions, nutritionists, and healthcare professionals develop effective dietary strategies and wellness programs.

Tools & Technologies Used:
1. Tableau – Data visualization and dashboard creation.
2. Microsoft Excel/CSV– Data preparation and preprocessing.
3. Data Analytics Techniques– Data cleaning, filtering, aggregation, and trend analysis
4.Stastical Analysis.

1.2-Purpose:

The purpose of the Comprehensive Analysis and Dietary Strategies project is to analyze students' food habits, dietary preferences, lifestyle patterns, and nutritional intake using data analytics and Tableau. The project aims to identify trends and relationships between eating behaviors, meal frequency, physical activity, Body Mass Index (BMI), and overall health. By transforming raw data into interactive dashboards and meaningful visualizations, it helps users understand key dietary patterns and make informed decisions. The insights generated support students, nutritionists, educational institutions, and health professionals in developing effective dietary strategies that encourage balanced nutrition, healthier food choices, and improved well-being. Ultimately, the project demonstrates how data-driven analysis can be used to promote better health outcomes and support informed decision-making in dietary planning.

- To understand the eating habits and nutritional behavior of college students.
- To identify factors influencing healthy and unhealthy food choices.
- To provide meaningful insights through interactive Tableau dashboards.
- To support students, educators, and health professionals in promoting healthier dietary practices.
- To encourage data-driven decision-making for nutrition and wellness programs.

2.IDEATION PHASE

The Ideation Phase of the Comprehensive Analysis and Dietary Strategies project focused on identifying the need for a data-driven approach to understand students' dietary habits and nutritional patterns. During this phase, the team analyzed the challenges faced by students in maintaining healthy eating habits and explored how data analytics could provide meaningful insights. Brainstorming sessions were conducted to define the project objectives, identify key stakeholders, determine the required data sources, and formulate business questions related to dietary preferences, meal frequency, BMI, physical activity, and nutrient intake. Based on these discussions, Tableau was selected as the visualization tool to create interactive dashboards that transform raw data into actionable insights. The ideation phase established a clear project roadmap for analyzing food choices and developing effective dietary strategies that support healthier lifestyles and informed decision-making.

2.1-Problem Statement:

Problem Statement 1: Health-Conscious College Student

I am	I'm trying to	But	Because	Which makes me feel
a health-conscious college student who wants to maintain a balanced diet while managing a busy academic schedule.	understand my daily food habits, nutritional intake, and identify healthier food choices.	my eating patterns are inconsistent, and I don't have a clear way to analyze my dietary habits and nutrition.	there is no centralized, interactive dashboard that provides visual insights into my food consumption and nutritional trends.	confused and uncertain, as I am unable to make informed decisions to improve my health and maintain a balanced lifestyle.

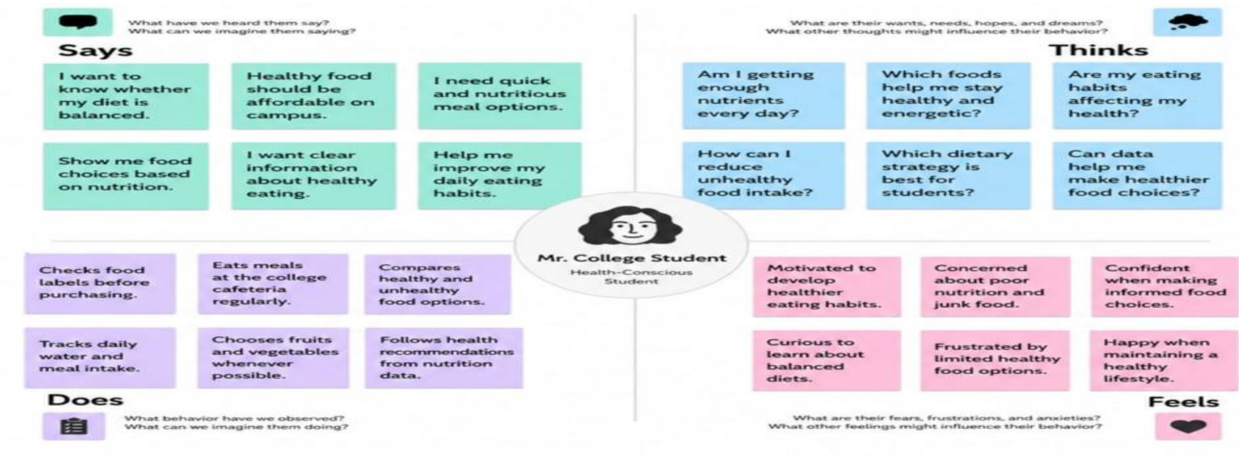
Problem Statement 2: Nutritionist/Diet Planner

I am	I'm trying to	But	Because	Which makes me feel
a nutritionist who helps individuals improve their eating habits through personalized dietary recommendations.	analyze food consumption patterns, nutritional intake, and lifestyle habits to recommend effective dietary strategies.	the data is scattered across multiple records and lacks clear visualizations for identifying nutrition trends.	there is no unified dashboard that combines dietary data, health indicators, and food consumption insights in one place.	limited and challenged, as I cannot efficiently provide data-driven dietary recommendations and monitor progress.

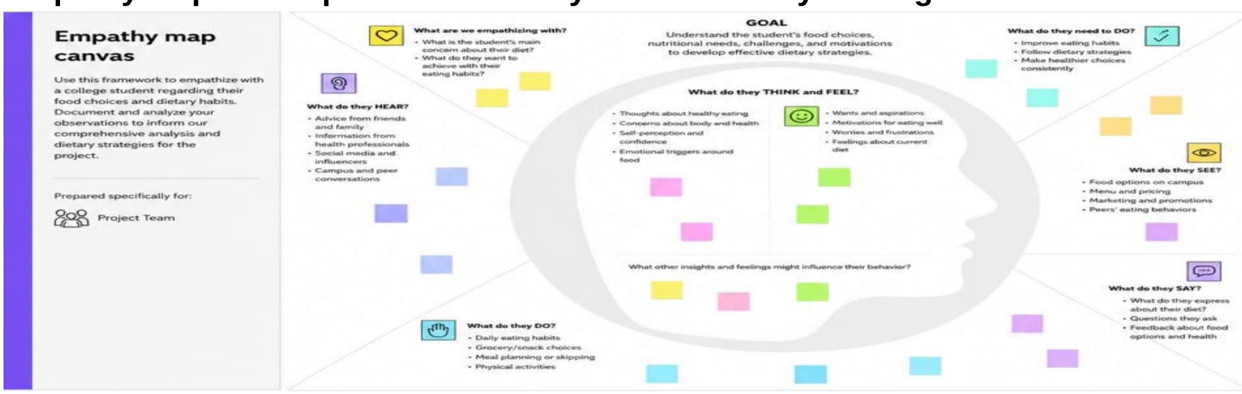
Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	I am a health-conscious college student who wants to maintain a balanced diet and improve my overall health.	I'm trying to track my daily food intake, analyze nutrition, and make healthier dietary choices.	But my dietary data is scattered, making it difficult to understand my eating habits and nutrition trends.	Because there is no centralized, interactive dashboard to visualize and analyze dietary information effectively.	Which makes me feel confused and uncertain about making informed dietary decisions and maintaining a healthy lifestyle.
PS-2	I am a nutritionist who helps individuals improve their eating habits through personalized dietary guidance.	I'm trying to analyze food consumption patterns and provide accurate dietary recommendations.	But the available nutrition data is scattered and lacks meaningful visualizations.	Because there is no unified dashboard to analyze dietary patterns and nutritional trends in one place.	Which makes me feel limited in providing quick, data-driven dietary recommendations.

2.2-Empathy Map Canvas :

Empathy Map-1:Comprehensive Analysis and Dietary Strategies

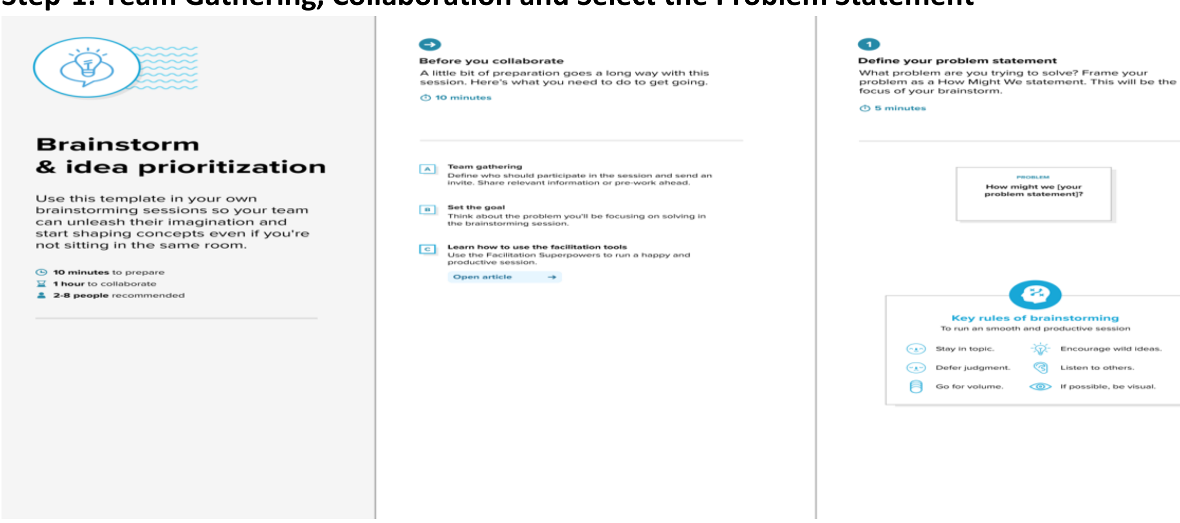


Empathy Map-2:Comprehensive Analysis and Dietary Strategies

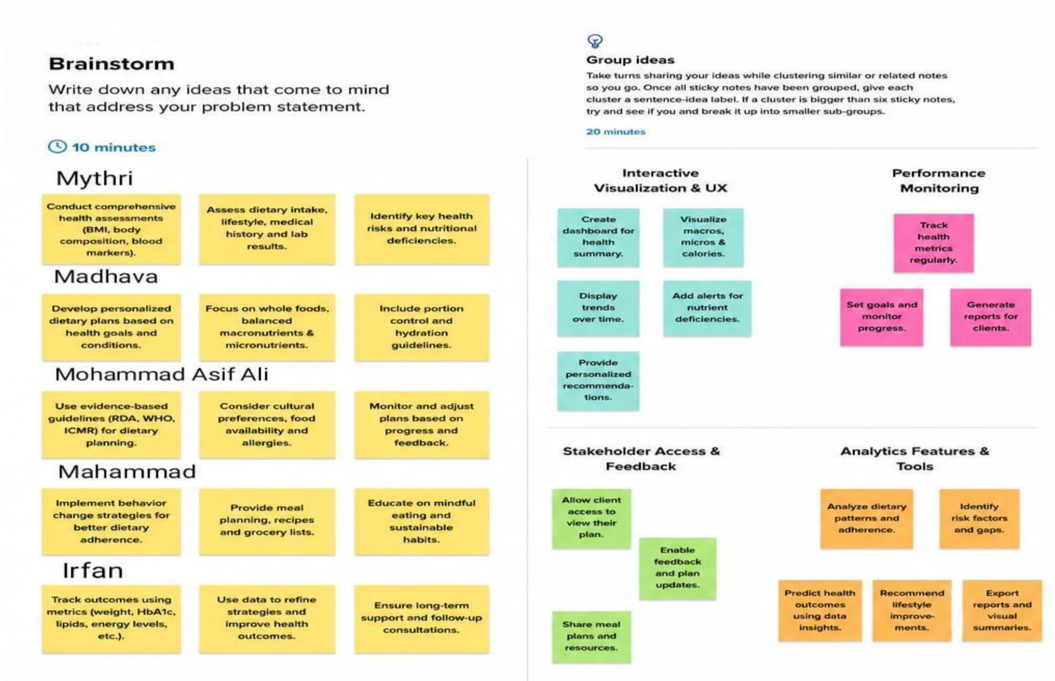


2.3- Brainstorming- Idea Generation- Prioritizaation Template:

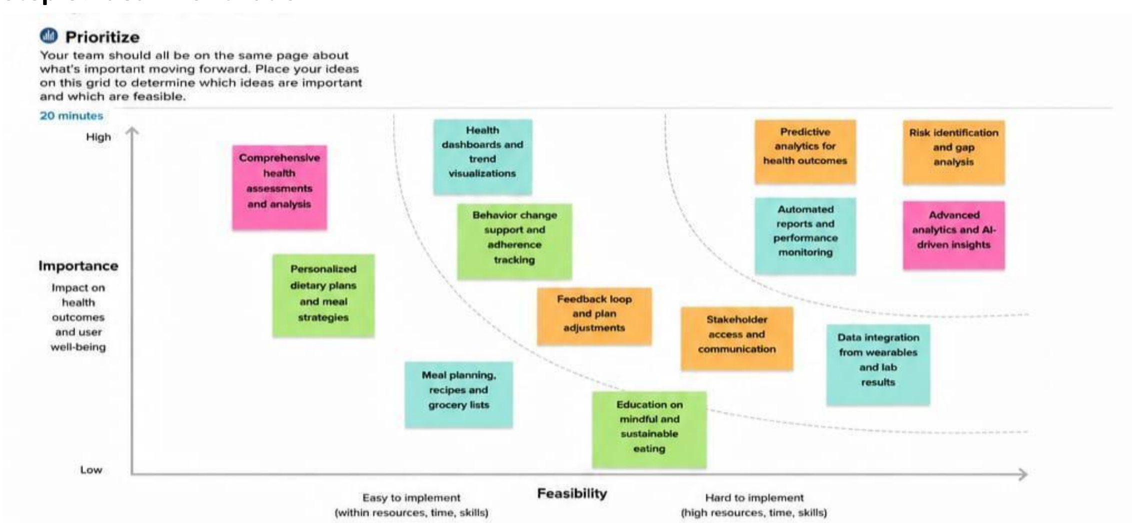
Step-1: Team Gathering, Collaboration and Select the Problem Statement



Step-2: Brainstorm, Idea Listing and Grouping



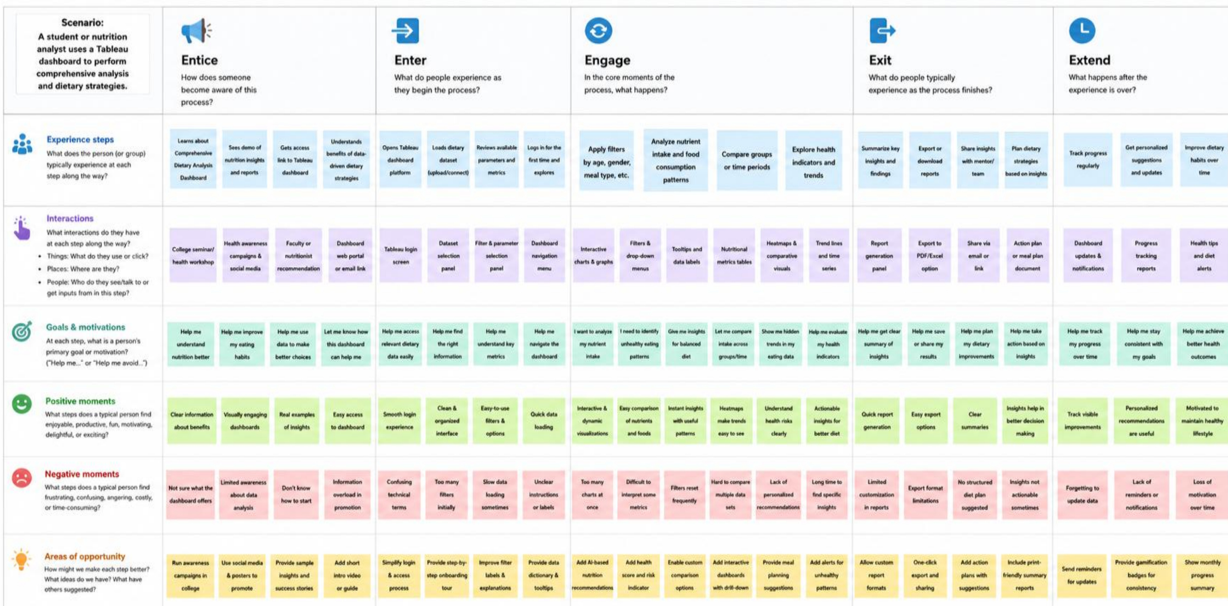
Step-3: Idea Prioritization



3-REQUIREMENT ANALYSIS:

The Requirement Analysis phase of the Comprehensive Analysis and Dietary Strategies project focused on identifying the data, tools, and objectives needed for the analysis. It involved collecting dietary and health-related data, defining key business questions, and determining the dashboard requirements. Tableau was selected for data visualization, while data cleaning and preparation ensured accurate analysis.

3.1-Customer Journey map:



3.2-Solution Requirement:

Functional Requirements:

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Collection	Collect dietary habits, meal frequency, breakfast consumption, calorie intake, exercise, health indicators, and demographic data from the College Food Choices dataset.
FR-2	Data Cleaning & Preparation	Remove duplicate records, handle missing values, standardize data types, and prepare the dataset for Tableau visualization.
FR-3	Data Analysis & Visualizations	Create interactive visualizations such as: 1. Gender vs Eating Habits 2. Breakfast Consumption Analysis 3. Daily Calorie Intake Distribution 4. BMI Category Analysis 5. Exercise Frequency vs Health Status 6. Fruit & Vegetable Consumption 7. Healthy Food Choice Factors 8. Stress Level vs Eating Behaviour
FR-4	Dashboard Development	Design an interactive Tableau dashboard with filters, charts, KPIs, and drill-down features for dietary analysis.

Non-functional Requirements:

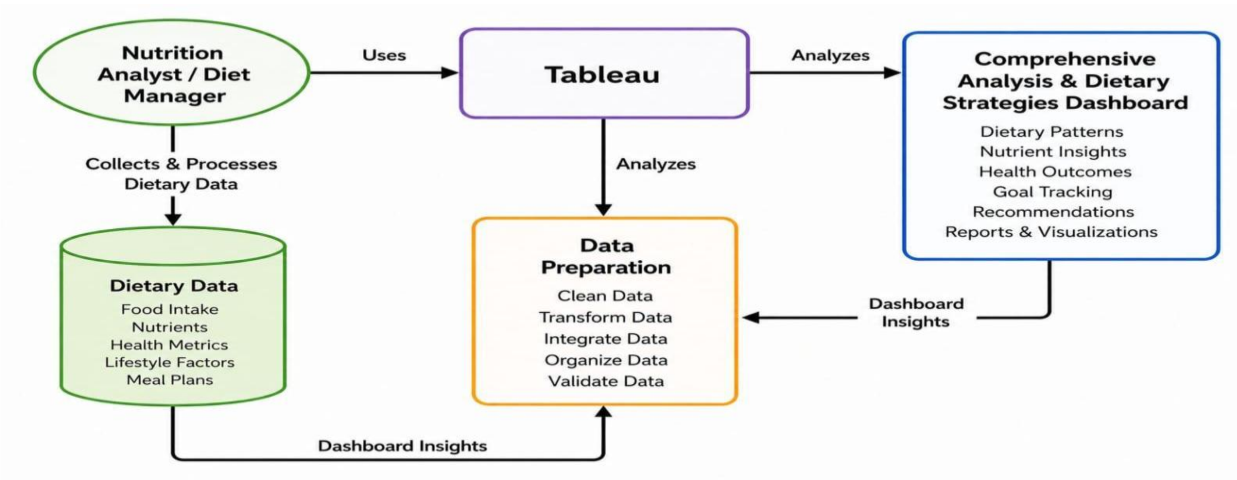
FR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	Dashboards should be simple, interactive, and easy to navigate using filters and charts.
NFR-2	**Security**	Ensure secure handling of health-related data with appropriate access controls.
NFR-3	**Reliability**	Dashboard should consistently provide accurate insights based on cleaned data
NFR-4	**Performance**	Dashboard should load quickly (within 5 seconds) and support smooth interaction.
NFR-5	**Availability**	Dashboard should be accessible whenever required with minimal downtime.
NFR-6	**Scalability**	The system should support larger dietary datasets and additional health parameters without affecting performance.

3.3-Data Flow Diagram:

Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

Example: (Simplified)



User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Data Analyst	Data Collection	USN1	Collect Dietary Dataset	Dietary data is successfully collected from all required sources.	High	Sprint 1
Data Analyst	Data Collection	USN2	Import Dataset into Tableau	The dataset is imported into Tableau without errors.	High	Sprint 1
Data Analyst	Data Preparation	USN3	Handle Missing Values	Missing values are identified and treated correctly.	High	Sprint 1
Data Analyst	Data Preparation	USN4	Create Calculated Fields	Required calculated fields (BMI,Calories,Nutrition Scores,etc.)are created.	High	Sprint 1
Data Analyst	Data Preparation	USN5	Clean Dietary Data	Duplicate and inconsistent dietary records are removed and standardized.	High	Sprint 1
Business User	Data Visualization	USN6	Create Nutritional Comparison Bar Chart	The bar chart correctly compares dietary and nutritional values.	Medium	Sprint 2

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Data Analyst	Data Collection	USN1	Collect Dietary Dataset	Dietary data is successfully collected from all required sources.	High	Sprint 1
Business User	Data Visualization	USN7	Create Food Category Pie Chart	Pie chart correctly displays food category distribution	Medium	Sprint 2
Business User	Data Visualization	USN8	Create Dietary Trend Line Chart	The line chart accurately displays dietary trends over time.	Medium	Sprint 2
Business User	Data Visualization	USN9	Create Demographic Analysis Dashboard	Dashboard correctly displays dietary patterns by age,gender or region.	Medium	Sprint 2
Business User	Dashboard Development	USN10	Develop Interactive Nutrition Dashboard	Interactive dashboard with filters,KPIs,and visualizations is completed successfully.	High	Sprint 2
Business User	Story Presentation	USN11	Create Tableau Story	Tableau Story presents dietary insights in a logical and meaningful sequence.	High	Sprint 2

3.4-Technology Stack:

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2.

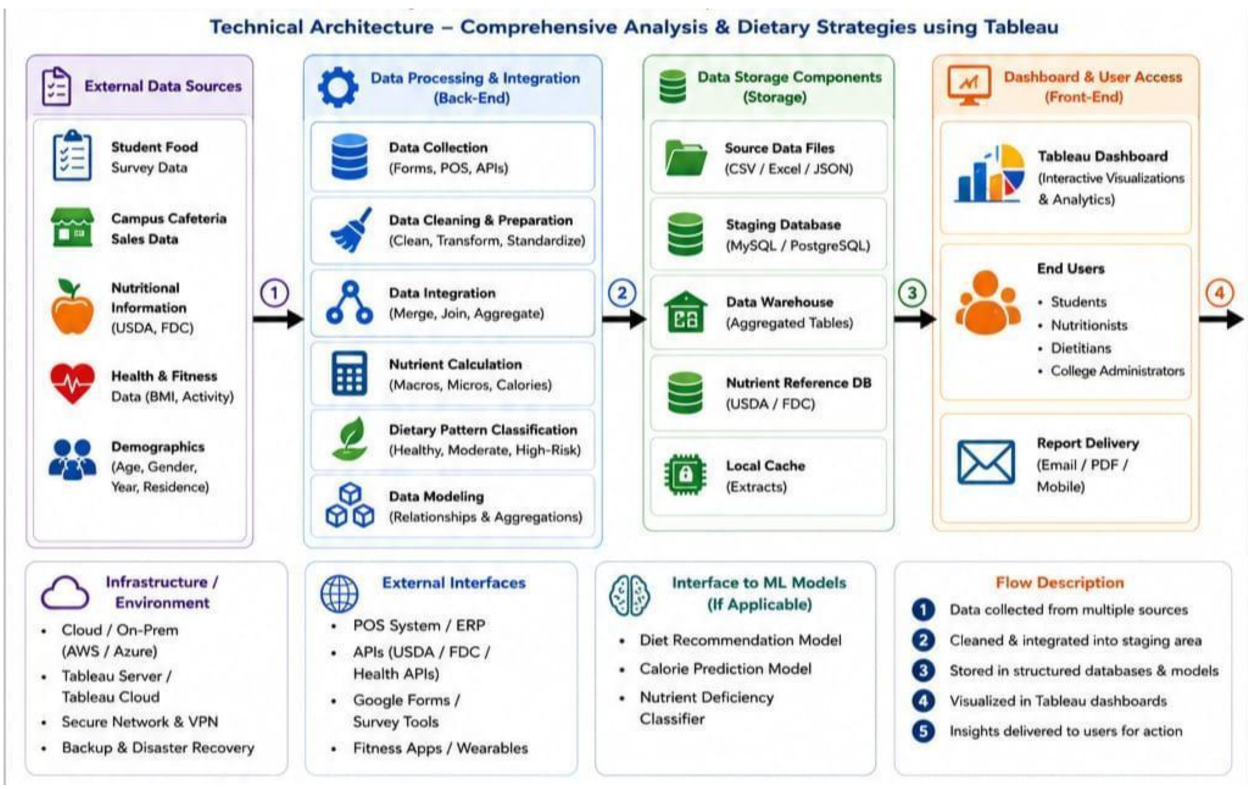


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	Data Source	Food survey responses, cafeteria sales, nutrition databases, health & demographic data.	Google Forms / POS / USDA API / CSV / Excel.
2.	Data Preparation	Cleaning, transforming and loading data for analysis.	Python (Pandas) / Tableau Prep / SQL.
3.	Data Visualization Tool	Creating interactive dashboards and insights.	Tableau Desktop / Tableau Online.
4.	User Interface	Web-based dashboards for different user roles.	Tableau Public / Embedded Tableau.
5.	Access Control	Manage logins and role-based access.	Tableau Server / Active Directory.
6.	Report Sharing	Share dashboards and reports with stakeholders.	Tableau Server / PDF / Email / Mobile.
7.	Feedback Collection	Collect user feedback for dietary improvements.	Google Forms / Microsoft Forms.

Table-2: Application Characteristics:

S.NO	Characteristics	Description	Technology
1.	Usability	User-friendly dashboards with filters, tooltips and clear KPIs.	Tableau UX/UI best practices.
2.	Performance	Fast data load and responsive interactions.	Tableau Extracts (TDE/Hyper) / Optimized SQL.
3.	Security	Role-based access and secure handling of health data.	Tableau Server Permissions / HTTPS / Encryption.
4.	Scalability	Handle large datasets and increase the user base.	Tableau Server / Cloud Storage.
5.	Availability	Dashboard accessible 24/7 with minimal downtime.	Cloud Hosting (AWS/Azure) / Tableau Server.
6.	Integration	Connect with multiple data sources and external systems.	Tableau Connectors / APIs.
7.	Real-time Data Refresh	Keep dashboards updated with latest food & health data.	Tableau Scheduler

4-PROJECT DESIGN:

The Project Design phase of the Comprehensive Analysis and Dietary Strategies project focused on planning the overall structure and workflow of the solution. It involved designing the data model, selecting appropriate charts and visualizations, and organizing interactive dashboards in Tableau. The dashboards were planned to display key insights such as dietary preferences, meal frequency, BMI, calorie intake, and physical activity. This phase ensured that the project was user-friendly, visually appealing, and capable of providing clear and meaningful insights for better dietary decision-making.

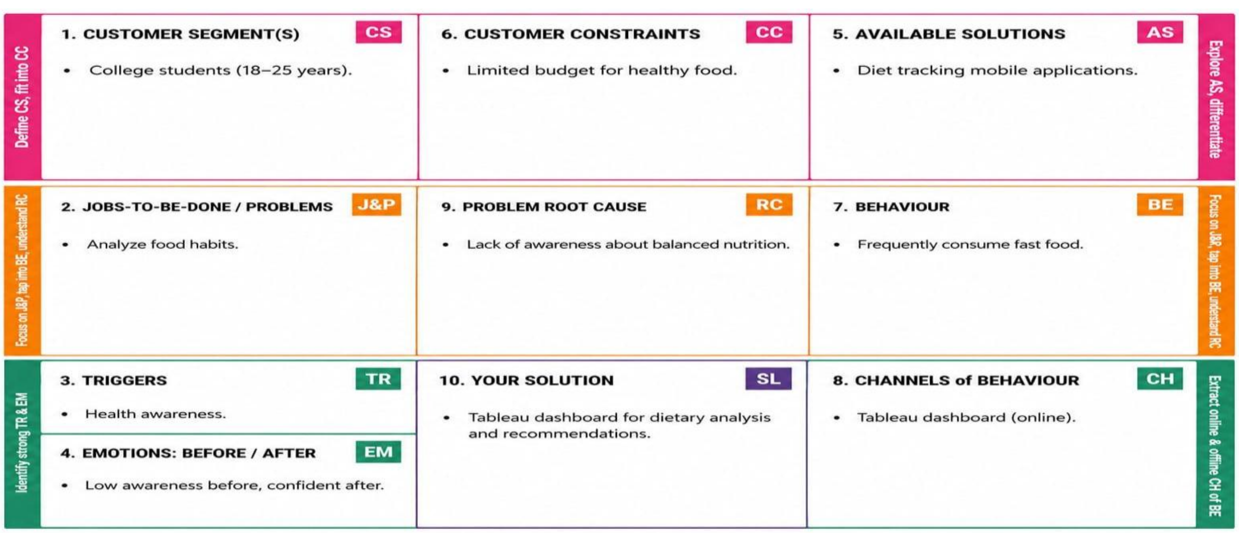
Problem – Solution Fit Template:

The Problem-Solution Fit simply means that you have found a problem with your customer and that the solution you have realized for it actually solves the customer's problem. It helps entrepreneurs, marketers and corporate innovators identify behavioral patterns and recognize what would work and why.

Purpose:

- ❑ Solve complex problems in a way that fits the state of your customers.
- ❑ Succeed faster and increase your solution adoption by tapping into existing mediums and channels of behavior.
- ❑ Sharpen your communication and marketing strategy with the right triggers and messaging.
- ❑ Increase touch-points with your company by finding the right problem-behavior fit and building trust by solving frequent annoyances, or urgent or costly problems.
- ❑ Understand the existing situation in order to improve it for your target group.

Template:



4.2-Proposed Solution:

Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Many individuals struggle to maintain healthy eating habits due to a lack of nutritional awareness and personalized dietary guidance. Poor food choices increase the risk of obesity, malnutrition, and lifestyle diseases. This project aims to analyze dietary patterns and recommend effective nutritional strategies using data analytics and visualization.
2.	Idea / Solution description	Analyze dietary habits, nutrient intake, food preferences, BMI, and lifestyle factors using Tableau dashboards. Develop interactive visualizations to identify nutritional gaps, compare eating patterns, and recommend personalized dietary strategies for healthier living.
3.	Novelty / Uniqueness	The solution integrates dietary data, nutritional analysis, BMI, calorie intake, and food preferences into a single interactive dashboard. It provides data-driven insights and personalized recommendations to improve dietary habits and overall health.
4.	Social Impact / Customer Satisfaction	The solution promotes healthier eating habits, increases nutritional awareness, reduces the risk of lifestyle diseases, and helps individuals make informed food choices, improving overall well-being and satisfaction.
5.	Business Model (Revenue Model)	The dashboard can be offered to hospitals, nutrition clinics, fitness centers, educational institutions, and wellness organizations through subscription-based analytics services, consultation packages, or customized health reports.
6.	Scalability of the Solution	The solution can be expanded for schools, colleges, hospitals, corporate wellness programs, fitness centers, and public health organizations. It can also be adapted for different age groups, dietary preferences, and regional food habits.

4.3-Solution Architecture:

Solution Architecture:

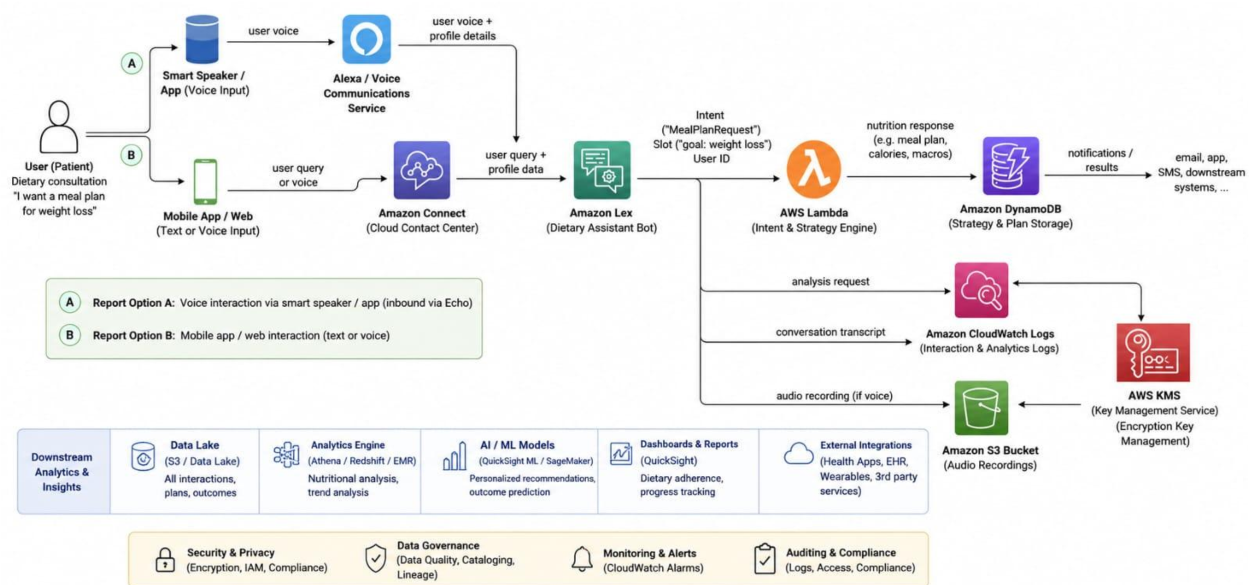
Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

● Find the best tech solution to solve existing business problems.

● Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.

● Define features, development phases, and solution requirements.

● Provide specifications according to which the solution is defined, managed, and delivered.



5-PROJECT PLANNING & SCHEDULING:

Project planning begins by defining the project objectives and business requirements. Relevant data sources are identified, collected, and prepared for analysis.A timeline is created for data cleaning, dashboard development, testing, and deployment.Roles, responsibilities, and resources are allocated to ensure smooth project execution.The project is reviewed regularly to ensure it meets deadlines and delivers meaningful insights.

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Data Collection	USN-1	As a member,I have gathered the data that is required to do the project.	2	High	Koneti Venkata Madhava
Sprint-1		USN-2	As a member, I have already gathered the data. so,I also performed loading that data.	1	Medium	Koneti Venkata Madhava
Sprint-1	Data Preparation	USN-3	As a member, I have handled the missing values.	3	Low	Koneti Venkata Madhava
Sprint-1		USN-4	As a member,I have created fields as per the requirement.	3	Medium	Koneti Venkata Madhava
Sprint-1		USN-5	As a member, I have handled the inconsistency in the data.	3	High	Koneti Venkata Madhava
Sprint-2	Data Visualization	USN-6	As a member, I have created different data visualizations like bar charts.	2	Medium	Mohammed Asif Ali Khan Gouri Patan
Sprint-2		USN-7	As a member, I have created different data visualizations like pie charts.	2	Medium	Mohammed Asif Ali Khan Gouri Patan
Sprint-2		USN-8	As a member, I have created different data visualizations like line charts.	2	Medium	Mohammed Asif Ali Khan Gouri Patan
Sprint-2		USN-9	As a member, I have created different data visualizations like map,bar chart,etc..,	5	Medium	Mohammed Asif Ali Khan Gouri Patan
Sprint-2	Dashboard	USN-10	As a member, I have developed a dashboard.	5	High	Mahammad Shaik
Sprint-2	Story	USN-11	As a member, I have developed a story.	5	High	Mahammad Shaik

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	20	6 Days	24 Jun 2026	29 Jun 2026	12	27 Jun 2026
Sprint-2	20	6 Days	30 Jun 2026	05 Jul 2026	21	01 Jul 2026

Velocity:

Imagine we have a 10-day sprint duration, and the velocity of the team is 20 (points per sprint). Let's calculate the team's average velocity (AV) per iteration unit (story points per day)

$$AV = \frac{sprint\ duration}{velocity} = \frac{20}{10} = 2$$

Burndown Chart:

A burn down chart is a graphical representation of work left to do versus time. It is often used in agile software development methodologies such as Scrum. However, burn down charts can be applied to any project containing measurable progress over time.

6-FUNCTIONAL AND PERFORMANCE TESTING:

Model Performance Testing:

Project team shall fill the following information in model performance testing template.

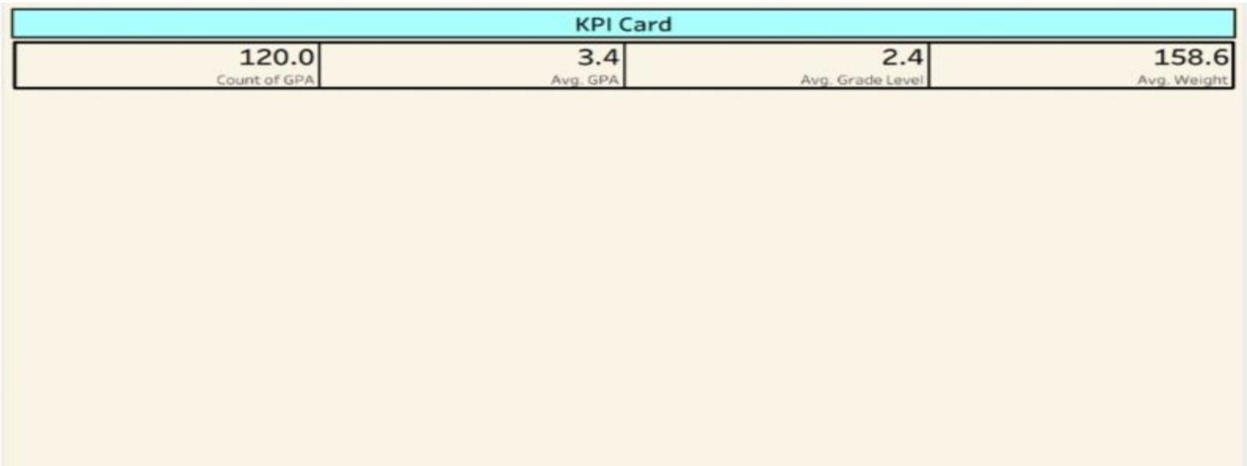
S.No.	Parameter	Screenshot / Values
1.	Data Rendered	Data of 120 Rows and 59 Columns has been rendered from a csv type file named as food_coded . The size of the file is 69 KB.
2.	Data Preprocessing	Handle missing values and remove duplicate values.
3.	Utilization of Filters	Measure Names, Cook and Gender: Male
4.	Calculation fields Used	Used 6 calculation fields. They are 1. Breakfast1 2. Gender1 3. Health 4. Calculation1 5. Food 6. Weight1
5.	Dashboard design	No of Visualizations / Graphs – 9 Visualizations 1.KPI Summary Card 2. Breakfast Consumption 3. Eating Out vs. Cooking Frequency 4. Health Category by Gender 5. Vitamins Consumption by Gender 6. Health vs. Greek Food Consumption 7. People Feeling Healthy and Life Rewarded 8. Weight vs. Exercise Frequency 9. Comfort Food Reasons

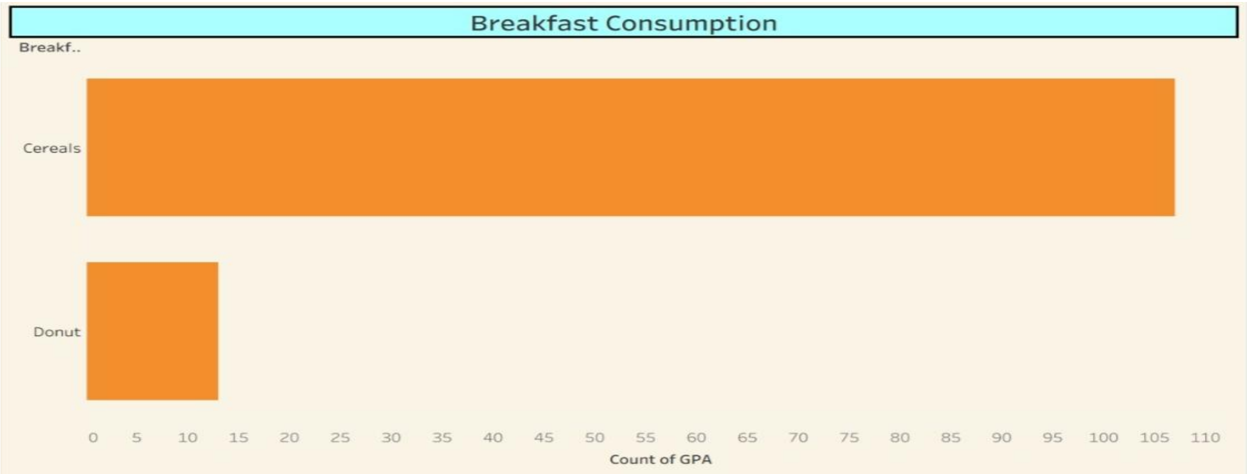
6	Story Design	No of Visualizations / Graphs -9 Visualizations 1.KPI Summary Card 2. Breakfast Consumption 3. Eating Out vs. Cooking Frequency 4. Health Category by Gender 5. Vitamins Consumption by Gender 6. Healtcoh vs. Greek Food Consumption 7. People Feeling Healthy and Life Rewarded 8. Weight vs. Exercise Frequency 9. Comfort Food Reasons

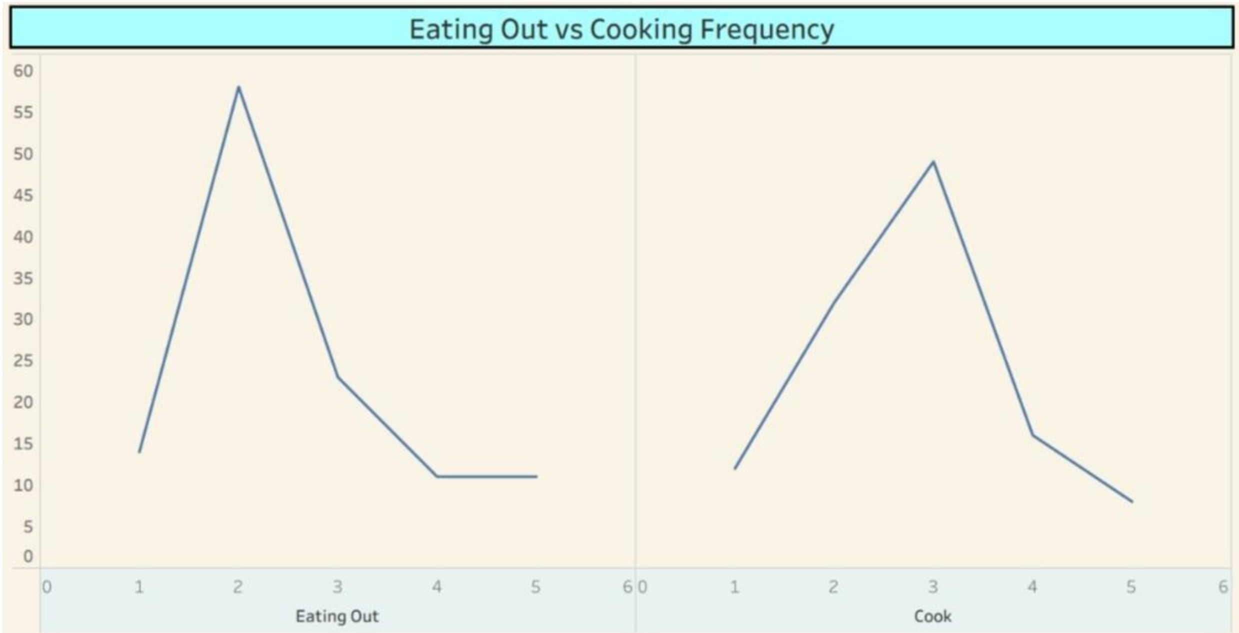
7.RESULTS:

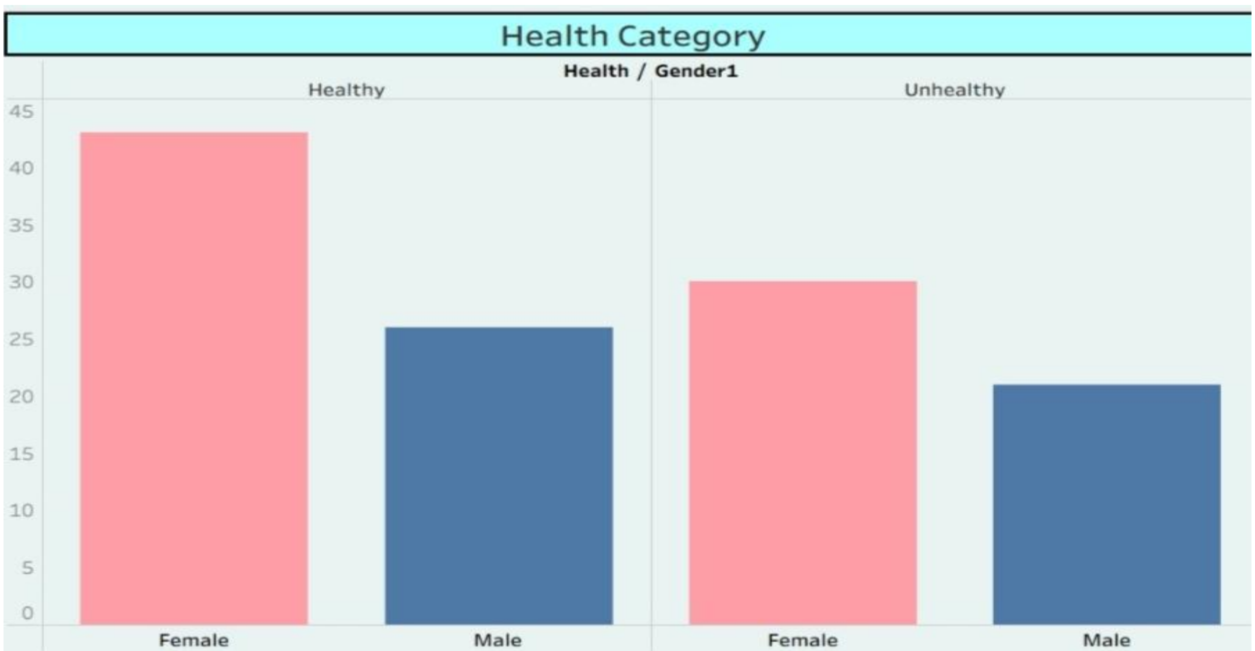
7.1 Output Screenshot:

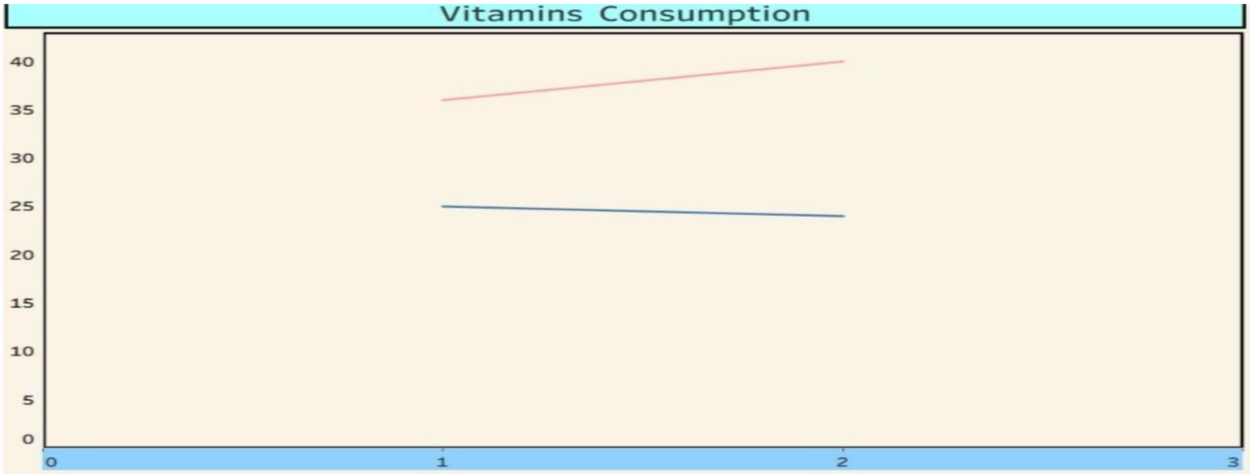
Comprehensive Analysis and Dietary Strategies:



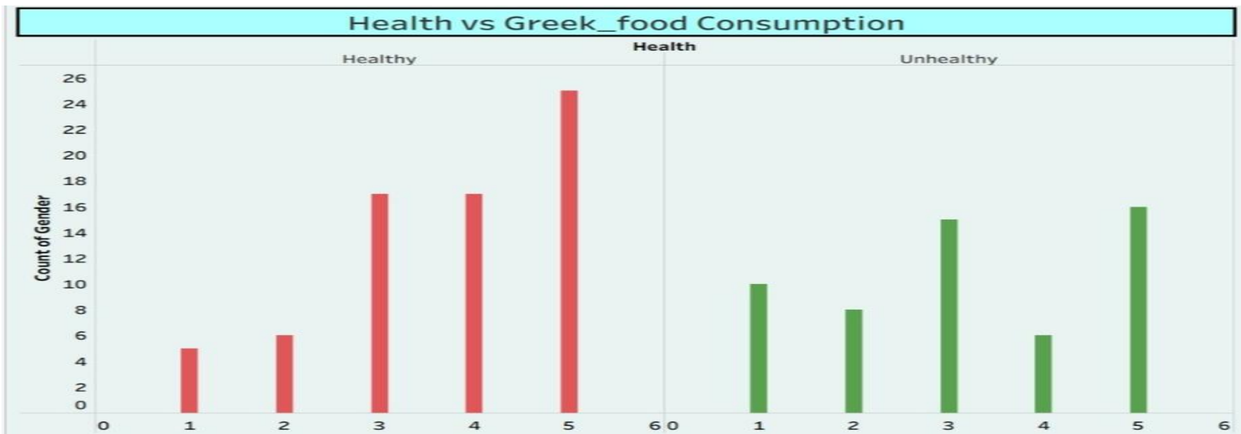



Eating Out vs Cooking Frequency
60
55
50
45
40
35
30
25
20
15
10
5
0
0
1
2
3
4
5
6
Eating Out
0
1
2
3
4
5
6
Cook

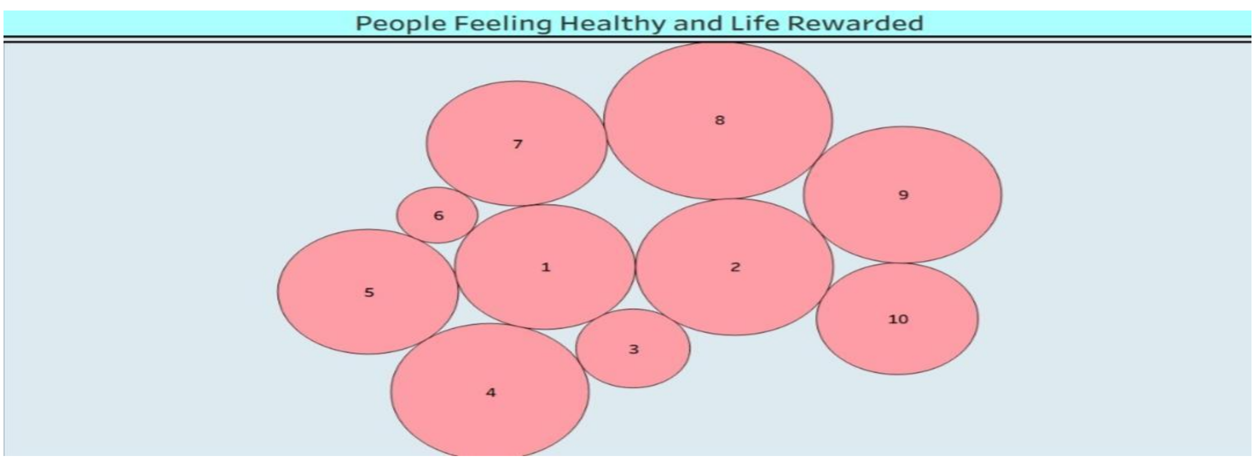

Health Category
Health / Gender1
Healthy
Unhealthy
45
40
35
30
25
20
15
10
5
0
Female
Male
Female
Male



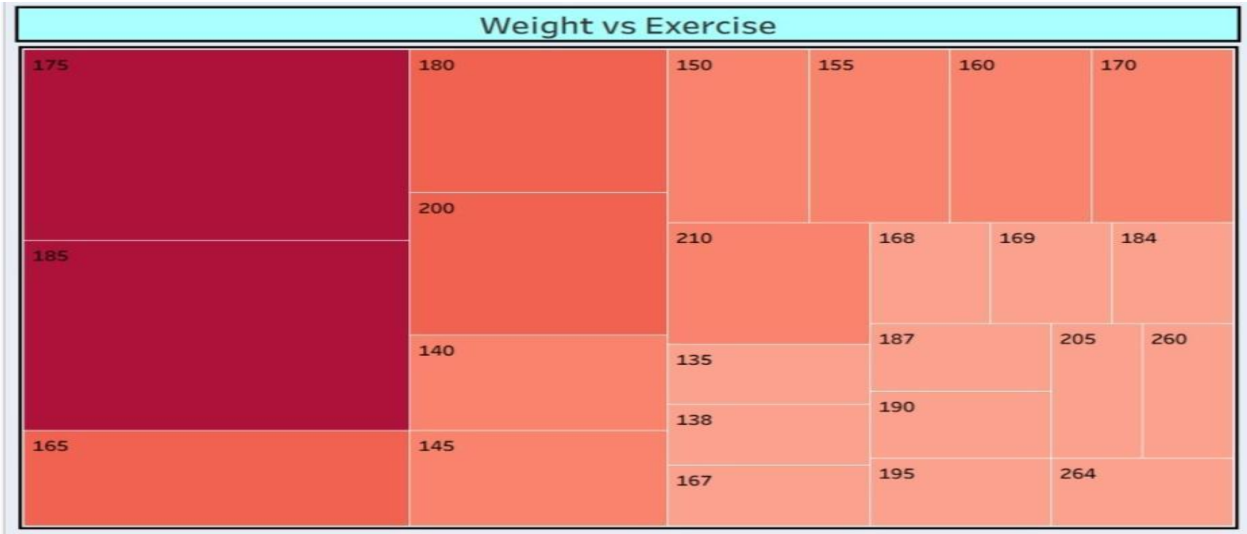
Vitamins Consumption
40
35
30
25
20
15
10
5
0
0
1
2
3

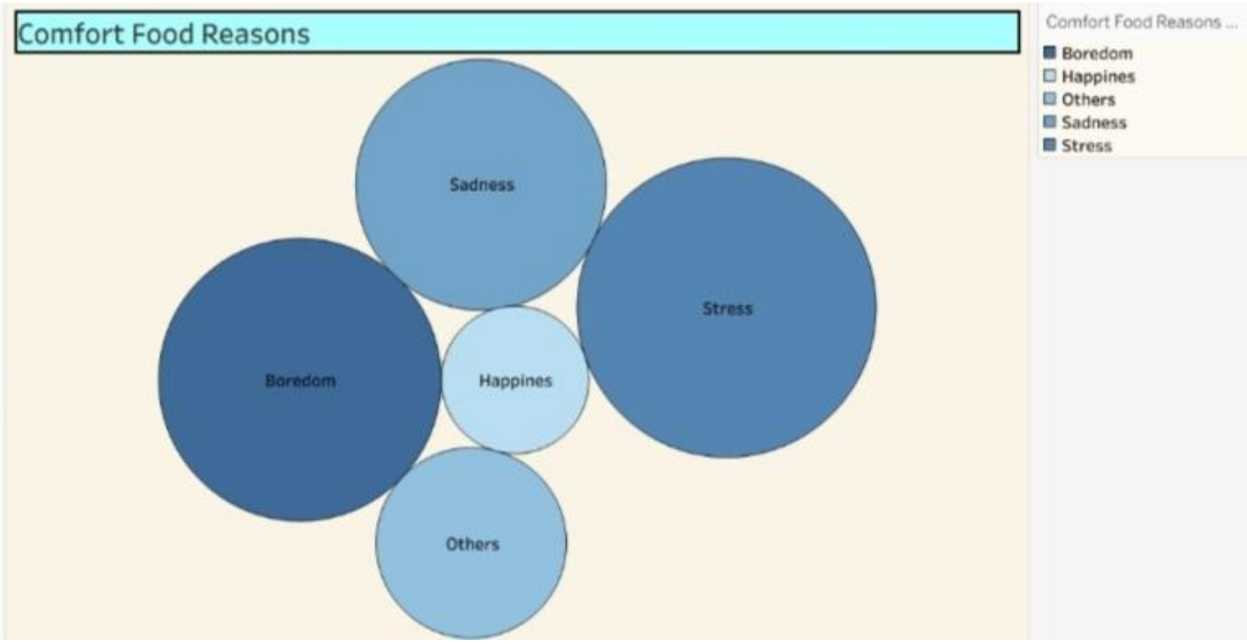


Health vs Greek_food Consumption
Health
Healthy
Unhealthy
Count of Gender
26
24
22
20
18
16
14
12
10
8
6
4
2
0
0
1
2
3
4
5
6

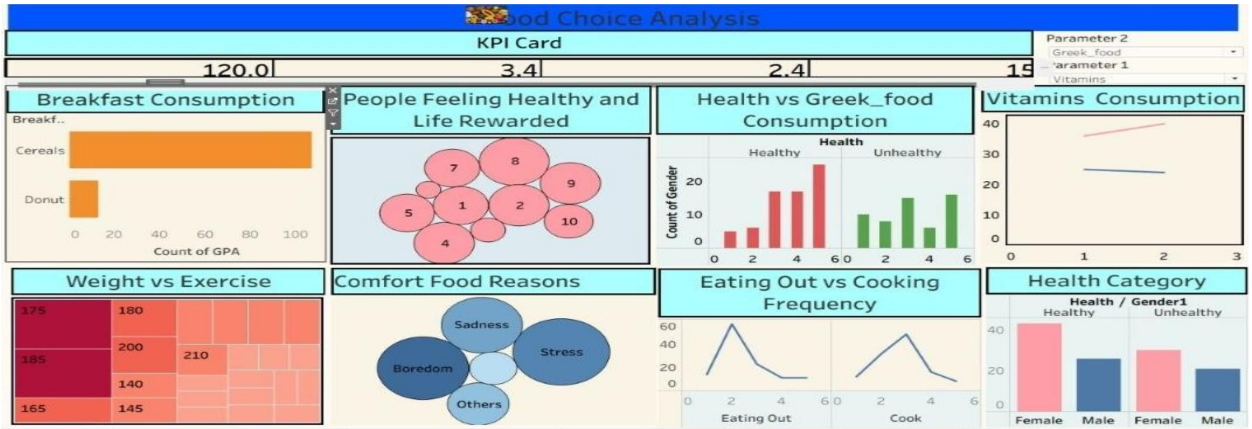


People Feeling Healthy and Life Rewarded
7
8
9
6
1
2
5
10
3
4

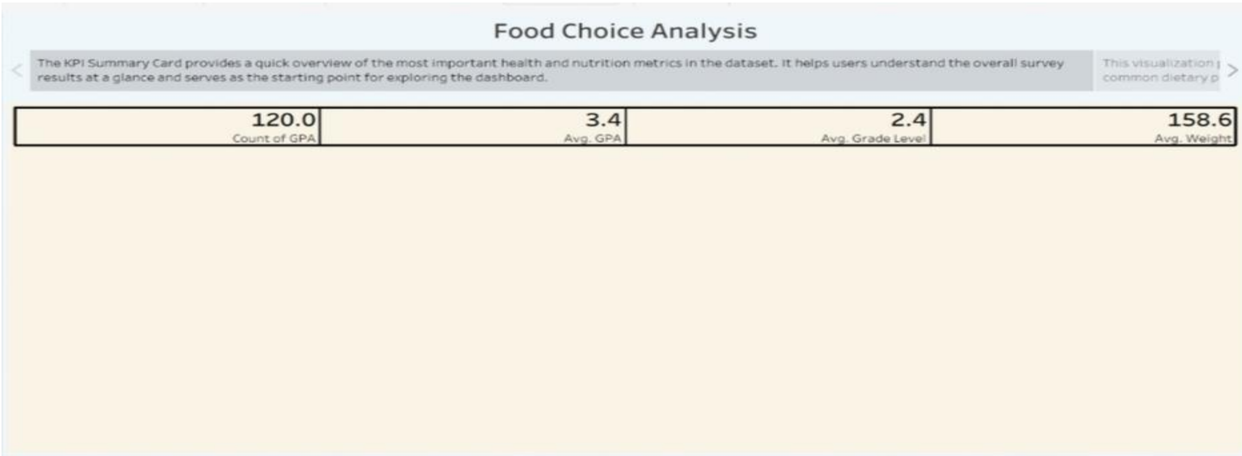

Weight vs Exercise
175
180
150
155
160
170
200
185
210
168
169
184
187
205
260
140
135
138
190
165
145
167
195
264


Comfort Food Reasons
Comfort Food Reasons ...
Boredom
Happines
Others
Sadness
Stress
Sadness
Stress
Boredom
Happines
Others

Dashboard:


Food Choice Analysis
KPI Card
120.0
3.4
2.4
Parameter 2
Greek_food
Parameter 1
Vitamins
Breakfast Consumption
Cereals
Donut
Count of GPA
People Feeling Healthy and Life Rewarded
Health vs Greek_food Consumption
Health
Healthy
Unhealthy
Count of Gender
Vitamins Consumption
Weight vs Exercise
Comfort Food Reasons
Eating Out vs Cooking Frequency
Eating Out
Cook
Health Category
Health / Gender1
Healthy
Unhealthy
Female
Male
Female
Male

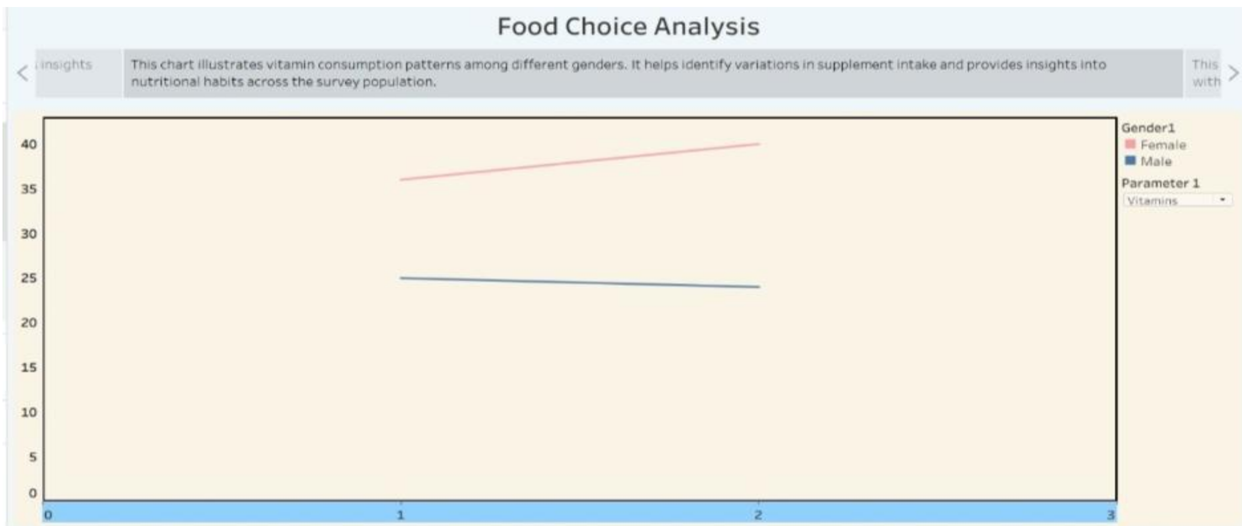
Story:


Food Choice Analysis
The KPI Summary Card provides a quick overview of the most important health and nutrition metrics in the dataset. It helps users understand the overall survey results at a glance and serves as the starting point for exploring the dashboard.
120.0
Count of GPA
3.4
Avg. GPA
2.4
Avg. Grade Level
158.6
Avg. Weight

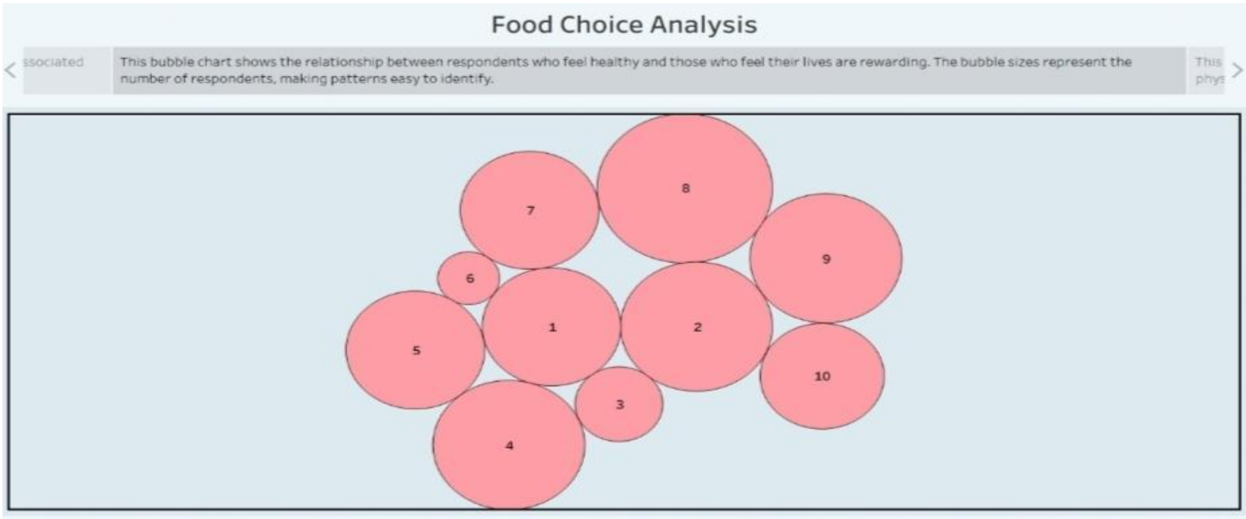

Food Choice Analysis
This visualization presents the breakfast consumption habits of respondents. It highlights how frequently participants eat breakfast, making it easy to identify common dietary patterns and compare eating behaviors.
Breakf..
Cereals
Donut
0 5 10 15 20 25 30 35 40 45 50 55 60 65 70 75 80 85 90 95 100 105 110
Count of GPA


Food Choice Analysis
This chart compares how often respondents eat out versus cook meals at home. It helps identify lifestyle preferences and reveals trends in daily food preparation and dining habits.
70 60 50 40 30 20 10 0 -10
-2 -1 0 1 2 3 4 5 6
Eating Out
0 1 2 3 4 5 6
Cook

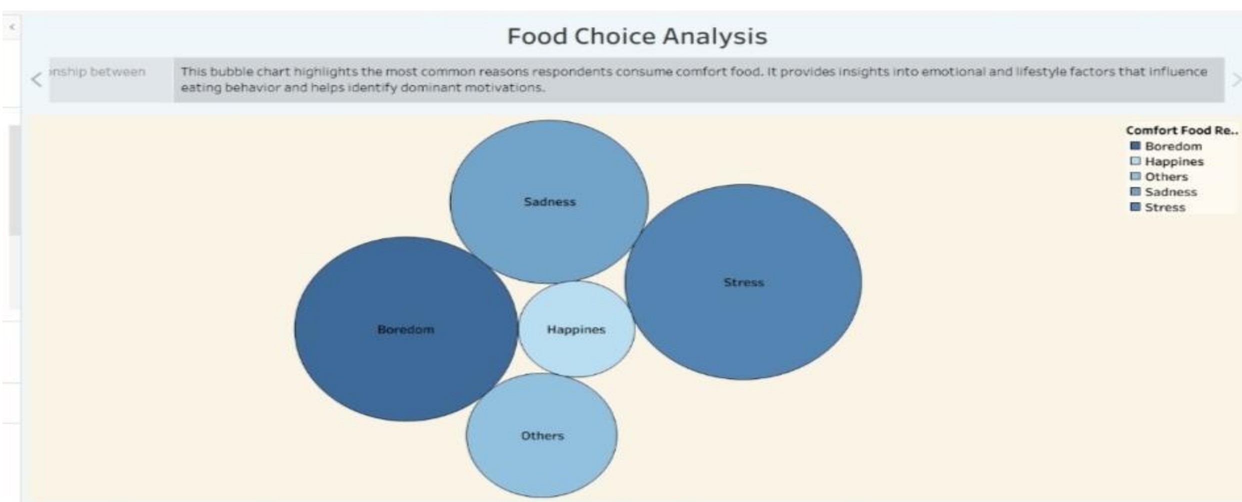

Food Choice Analysis
This visualization compares health categories across male and female respondents. It highlights gender-based differences in health status and provides insights into overall health distribution.
Health / Gender1
Healthy
Unhealthy
45
40
35
30
25
20
15
10
5
0
Female
Male
Female
Male
Gender1
Female
Male


Food Choice Analysis
This chart illustrates vitamin consumption patterns among different genders. It helps identify variations in supplement intake and provides insights into nutritional habits across the survey population.
40
35
30
25
20
15
10
5
0
0
1
2
3
Gender1
Female
Male
Parameter 1
Vitamins


Food Choice Analysis
This visualization explores the relationship between health categories and Greek food consumption. It helps determine whether food preferences are associated with respondents' perceived health status.
Health
Healthy
Unhealthy
Count of Gender
26
24
22
20
18
16
14
12
10
8
6
4
2
0
0
1
2
3
4
5
6
0
1
2
3
4
5
6
Health
Healthy
Unhealthy
Parameter 2
Greek_food


Food Choice Analysis
This bubble chart shows the relationship between respondents who feel healthy and those who feel their lives are rewarding. The bubble sizes represent the number of respondents, making patterns easy to identify.
8
7
9
6
1
2
5
10
3
4


Food Choice Analysis
This treemap visualizes exercise frequency across different weight groups. It helps identify activity patterns and provides insights into the relationship between physical activity and body weight.
Count of Exercise
0
6
175
165
140
160
170
210
145
180
135
169
184
187
185
150
138
190
205
260
200
167
155
195
168
264


Food Choice Analysis
This bubble chart highlights the most common reasons respondents consume comfort food. It provides insights into emotional and lifestyle factors that influence eating behavior and helps identify dominant motivations.
Comfort Food Re..
Boredom
Happines
Others
Sadness
Stress
Sadness
Stress
Boredom
Happines
Others

8.ADVANTAGES & DISADVANTAGES:

Advantages

- Provides clear and interactive visual analysis of dietary data.
- Helps identify unhealthy eating patterns and nutritional gaps.
- Supports better dietary planning and health awareness.
- Enables quick comparison of multiple health and food-related factors.

Disadvantages

- Results depend on the accuracy and completeness of the collected data.
- Self-reported food habits may introduce bias.
- The study may not represent all colleges or regions.
- Dietary habits can change over time, requiring regular data updates.

9.CONCLUSION:

The project successfully analyzes college students' dietary habits using Tableau.Interactive dashboards help identify key trends in food choices, nutrition, and lifestyle.The findings support healthier dietary strategies and informed decision-making.This analysis can help educational institutions encourage better eating habits and improve student wellness.Overall, the project demonstrates how data visualization can transform dietary data into meaningful insights for health improvement.

10.FUTURE SCOPE:

The future scope of Comprehensive Analysis and Dietary Strategies is highly promising as data analytics continues to transform the healthcare and nutrition sectors. Future versions of the project can incorporate real-time health data from wearable devices and mobile health applications to provide more accurate and personalized dietary recommendations. By integrating Artificial Intelligence (AI) and Machine Learning (ML), the system can predict nutritional deficiencies, identify health risks, and recommend customized meal plans based on individual health conditions and lifestyle patterns. The project can also be expanded to include larger datasets from hospitals, fitness centers, and public health organizations, enabling more comprehensive analysis. Additionally, interactive dashboards can support nutritionists, healthcare professionals, educational institutions, and policymakers in making data-driven decisions to improve public health, promote healthy eating habits, and reduce the risk of lifestyle-related diseases.

11.APPENDIX:

Dataset link:

https://drive.google.com/file/d/1Q-Vjv-jVbmI8wj42Ew4ocYj90GTyHRM3/view?usp=sharing

Github link:

https://github.com/saimythriveluru13/Comprehensive-Analysis-and-Dietary-Strategies-with-Tableau-A-College-Food-Choices-Case-Study

Project demo link:

https://drive.google.com/file/d/1sVTgWkl7PGwLZmN7SBL3kswQWZbNk_EC/view?usp=sharing